INDIA.RUNS

Build what next India runs on

Track 1 AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

- Extend existing Redrob capabilities
- Improve experiences for Redrob users
- Introduce new AI-native workflows
- Strengthen ecosystem engagement and network effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

- How the idea leverages existing Redrob capabilities
- What new capability, workflow, or value layer is introduced
- How the idea strengthens the overall Redrob ecosystem
- Why Redrob is uniquely positioned to enable this experience

Team Name : peter14l

Problem Statement : Challenge 1 — The AI Systems Architect: Reimagining Work

Team Members : Peter (Solo Participant)

Problem Definition

Questions

- . Traditional recruitment is broken — keyword filters miss 70% qualified candidates. Screenings slow, biased, and ignores India's multilingual talent across 22+ languages.

. Who experiences this?

. Recruiters drowning in 500+ apps/role. HR wasting 60% time on manual screening. Candidates from Tier 2/3 cities systematically excluded.

. Why is current approach insufficient?

. ATS tech 10+ yrs old. Black-box AI rejects without explanation. No multilingual/voice support. Linear filtering can't handle scale.

Opportunity & Vision

Questions

- **. Why is this important?**

. $2B+ India recruitment TAM, 15% CAGR. 1.5M+ engineers/year, mostly from Tier 2/3 cities. Redrob targeting 10-20M MAU by Dec 2026.

. What future state are you enabling?

. AI-native hiring where every candidate gets a fair shot regardless of language or city. Recruiters go from 5-7 day shortlists to ~10 minutes.

Solution Overview

Questions

- . What is your proposed solution?

. AI Recruiter Agent Swarm — 5 specialized AI agents collaborating autonomously: Orchestrator (routing), Sourcer (crawling), Matcher (semantic ranking), Screener (voice interviews), Scheduler (calendar coordination). Full pipeline in ~10 min.

. What makes it AI-native?

. Agents make autonomous decisions via LLM reasoning. Voice screening adapts dynamically. Semantic matching understands context/transferable skills. Orchestrator uses state machine with 12 escalation rules for human handoff.

. Which Redrob capability does it build upon?

. Redrob's 15M+ job listings, 30M+ professional profiles, 30+ language infrastructure. Integrates as intelligent hiring brain, making Redrob the AI-native hiring destination.

User Journey / Workflow Diagram

Questions

- . Recruiter submits JD → Orchestrator parses + activates swarm → Sourcer searches LinkedIn/GitHub/Naukri/Redrob in parallel → Matcher embeds + scores (5 dimensions) → Screener calls top candidates in Hinglish → Scheduler negotiates slots → Recruiter reviews shortlist with full reasoning.

. Information flows via async message bus (sender, recipient, type, payload, timestamp). Every decision logged and auditable. Supports pause/resume/rollback.
. Integrates with Redrob dashboard, profile database, and multilingual infrastructure.

Mandatory Visual

- Workflow Diagram

AI Logic & Decision Flow

Questions

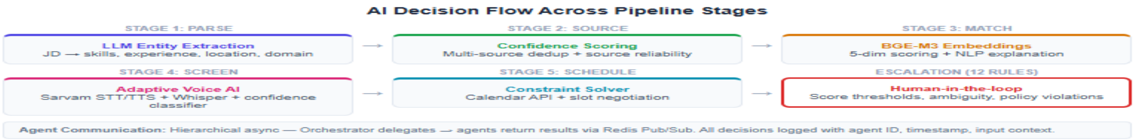
- . Every pipeline stage: JD parsing via LLM → semantic embedding scoring → adaptive question generation → voice response analysis (fluency, confidence) → scheduling optimization. Orchestrator decides human escalation via 12 rules.

. Matcher computes 5-dimension scores (skills, experience, location, domain, transferable) per candidate with NLP explanations. Orchestrator aggregates with threshold + LLM judgment.

. Hierarchical swarm: Orchestrator delegates, agents return results via Redis Pub/Sub. Each agent is an independent microservice.

Mandatory Visual

- AI Flow Diagram



System Architecture

Questions

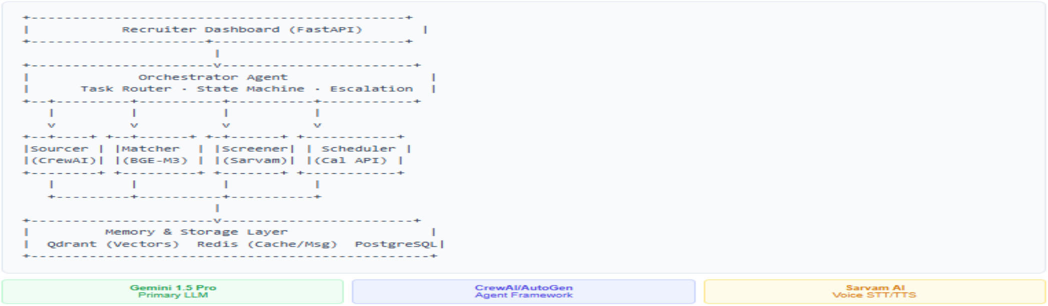
- . Orchestrator (state machine, 5 phases), Sourcer (multi-source crawler), Matcher (BGE-M3 + 5D scorer), Screener (Sarvam AI STT/TTS), Scheduler (calendar + WhatsApp). Storage: Qdrant (vectors), Redis (cache/msg), PostgreSQL.

. Redis Pub/Sub for async comms. FastAPI serves dashboard. Each agent independently deployable on Kubernetes.
. Leverages Redrob profile DB, job platform, auth/sessions, and multilingual infra.

Mandatory Visual

- Architecture Diagram

System Architecture Diagram



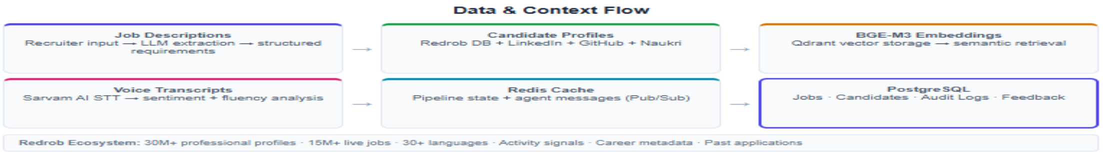
Data, Context & Intelligence Layer

Questions

- . JDs from recruiters, candidate profiles (Redrob/LinkedIn/GitHub), voice interview transcripts, recruiter feedback. Enriched with activity signals and career metadata.

. BGE-M3 embeddings in Qdrant for semantic search. Redis for pipeline state + message bus. PostgreSQL for structured persistence (jobs, candidates, audit logs).
. Redrob’s 30M+ profiles + 15M+ jobs + 30+ languages provide foundational data. User activity signals enrich matching beyond single-source profiles.

Mandatory Visual



- Data Flow Diagram

Scalability & Technical Feasibility

Questions

- **. Implementation:**

. FastAPI microservices on K8s. CrewAI/AutoGen framework. Gemini 1.5 Pro LLM. Sarvam AI voice. Qdrant vectors. Redis cache/msg. PostgreSQL storage.

. How does it scale?

. Each agent independently scalable. Qdrant for million-scale vectors. Redis cluster for queuing. Horizontal scaling per job volume.

. What challenges exist?

. Voice latency (<2s target), API rate limits (intelligent caching), data privacy (PII masking, encryption), LLM hallucination (confidence thresholds + human escalation).

Redrob Ecosystem Integration

Questions

- . Leverages: 15M+ jobs, 30M+ profiles, 30+ languages, auth/sessions, user activity signals.

. New capability: End-to-end autonomous hiring pipeline (JD to shortlist in ~10 min). Multi-source sourcing, semantic matching, voice screening in Indian languages.
. Ecosystem impact: Makes Redrob the intelligent hiring destination. Network effects: better matching → more candidates → more recruiters → more data → better matching.
. Future opportunities: Predictive hiring, offer negotiation, cultural fit prediction, skills gap analysis, API marketplace, white-label agents.

Mandatory Visual

- Ecosystem Integration Diagram

Impact & Success Metrics

Questions

- **. Expected outcomes:**

. Time to shortlist: 5-7 days → ~10 min (99.9% reduction)
. Recruiter screening: 60% → 10% of day (5x efficiency)
. Candidates evaluated: 50-100 → 500+ per role
. Cost-per-hire: ~70% reduction for mid-senior roles
. Candidate diversity: +40% from language-inclusive screening
. Candidate NPS: 22 → 60+ projected

. How to track success?

. Pipeline completion rate, time-per-phase, diversity metrics, recruiter/candidate satisfaction, escalation rate.

. Value created:

. Users: faster hiring, better matches, transparency. Redrob: differentiated AI capability, platform stickiness, network effects, per-job/per-candidate revenue.

Future Roadmap

Questions

- **2-3 year evolution:**

. Q4 2026 — MVP: 5 beta customers. LinkedIn sourcing, Matcher + Sourcer + Orchestrator.
. Q1 2027 — Growth: 5 agents in production. Hinglish voice screening. 50 customers.
. Q2 2027 — Platform: 6+ Indian languages. Feedback learning loop. 200+ customers.
. Q3 2027+ — Scale: Predictive hiring, offer automation, cultural fit prediction, retention scoring.

- **Future capabilities:**

. Skills gap analysis, career pathing, labor market intelligence, salary benchmarking, internal mobility marketplace.

- **Broader vision:**

. Redrob becomes India’s AI OS for work. Every job match runs through the intelligent pipeline. Hiring becomes a 10-min background process.

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THANK YOU